

GROUP NUMBER: 48

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AVAILABLE INPUTS: Input files are available in the hdfs file system: /data/BDC2526

TEST: The goal of this test is **to assess the scalability of your MRFairFFT program**. The test must be performed on the artificial datasets ArtificialNx DyKz.csv found on the hdfs file system. Conventionally, the name of the dataset represents some basic information: ArtificialNx DyKz.txt contains x points in R^y forming z natural clusters.

Use this table to report the results of the requested experiments. For all experiments use kA=8, kB=2, L=16, and filling only those cells for which the execution required at most 5 minutes.

Dataset	Runtime (2 executors)	Runtime (4 executors)	Runtime (8 executors)	Runtime (16 executors)	Obj. function (16 executors)
ArtificialN130kD7k10	829ms	448ms	328ms	178ms	38.8797281075 53674
ArtificialN1.3mkD7k10	2507ms	1370ms	737ms	442ms	38.9626983395 0417
ArtificialN13mD7k10	82358ms	39338ms	26529ms	13613ms	39.5448137118 1054
ArtificialN130mD7k10	OOM	OOM	OOM (With L = 512 81097ms)	OOM (With L = 128 135993ms)	OOM (With L = 128 39.7235425470 50066)

ANSWER THE FOLLOWING QUESTIONS

- Did you use some Generative AI tool? YES
- (Only if you answered YES to previous question) Describe in at most 10 lines how you used the Generative AI tool.

We used a Generative AI tool to generate some code to test our project and to better learn functional programming, for example we generated a simple function to plot the points and the centroids (only when they had 2 dimensions).